Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

A sample warms by 3°C after 500 J is transferred into it. **Explain** why the temperature rise and heat transfer need different units.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

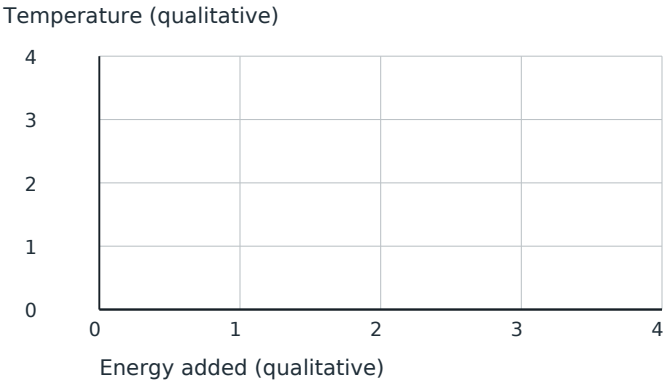
02 Worked example: explain the model

Reference: Q = mL; Q = mc ΔT.
Use mL during a phase change at constant temperature, and mc ΔT outside that interval. L is latent heat.

Worked reasoning: Melting 2 kg of a substance with latent heat 100 J/kg requires 200 J. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting.
Mark the plateau.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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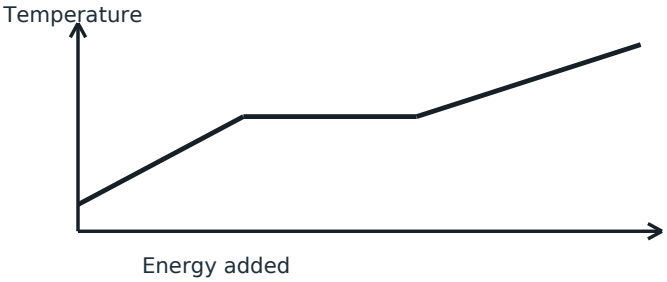
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the heating curve, identify the flat interval.
Explain why added energy does not raise temperature during that interval.

Sentence starter: My claim is … . The diagram/data show …, so … .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

04 Apply the physics

A phase-change simulation uses latent heat L = 200 J/kg. A 3 kg sample starts at its transition temperature.
Label the phase-change energy arrow and calculate the energy required to finish the transition.
Explain why this does not determine a temperature rise.

Sentence starter: My claim is … . The diagram/data show …, so … .

Initial T / phase: _____ Final T / phase: _____



Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student uses mc ΔT for a melting interval with ΔT=0 and concludes no energy is required.
Explain the missing term.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A spring has k=400 N/m and compression 0.06 m.
Find force magnitude and describe its direction.
Mark equilibrium, displacement and restoring-force direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg sample with L=150 J/kg changes phase completely.
Find Q.
Label the energy arrow and identify what changes in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Compare heat needed to melt 1 kg and 3 kg of the same substance.
Explain why temperature can remain unchanged in both.
Label the energy arrow and identify what changes in the sample.

Sentence starter: Both … . They differ in … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3